

Education 4.0 Approach for New Careers at Mexico City-IPN



Vladimir Avalos-Bravo , Jorge Toro González, and Yaravid García Pérez

Abstract Mexico City Instituto Politecnico Nacional (IPN) has carried out Academic Programs development that respond to public and private productive sector's needs, performing an evaluation process that allows curricular strengths and weaknesses identification. This process is an essential exercise that allows to systematically collect quantitative and qualitative data to support new careers creation with solid arguments for their realization. Every Academic Programs design consists of a deep analysis of similar careers in other national and international educational institutions, external references as every career basis, labor market analysis of each proposed topics to determine industry needs and verify their relevance. Subsequently, objectives, educational profile, mission, vision, and graduation profile are defined to rise the program curricular structure, defining learning units and their characteristics (number of theoretical and practical hours, credits), knowledge transfer, teaching strategies and other curricular elements. The design of the trajectories or curricular maps of the Upper Middle High Level programs, comprise learning units cataloged in three training areas: (1) Institutional training, (2) Basic scientific, humanistic, and technological training and (3) Vocational training, meanwhile Higher Education programs comprise learning units cataloged in four training areas: (1) Institutional training, (2) Basic scientific training, (3) Professional training and (4) Terminal and integration training. This paper shows a brief description of the new careers created in the 2018-2019 period, considering the most important characteristics, curricular trajectories with an Education 4.0 approach.

V. Avalos-Bravo (✉)

Instituto Politécnico Nacional, DEV, SEPI-ESIQIE, UPIEM, SARACS Research Group ESIME
Zacatenco, Mexico City, Mexico

e-mail: ravalos@ipn.mx

J. T. González

Instituto Politécnico Nacional, ESFM, Mexico City, Mexico

e-mail: jtoro@ipn.mx

Y. G. Pérez

Instituto Politécnico Nacional, ENCB, Mexico City, Mexico

e-mail: ygarciap1000@alumno.ipn.mx

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1 Introduction

In an era that focuses on smart technology, artificial intelligence, big data, data analytics, customizable services and robotics, educational institutions must produce a highly skilled and capable workforce who can take advantage of the tools available in this technologically transformed world. Education 4.0 requires promoting students' digital skills and abilities so that schools train graduates who can join the industry that already has many automated processes and are developing new technologies. Technological advances in different fields of knowledge and the immersion of information and technologies (IT) in all human activities, mark the concept of a new era, in which industries combine both physical, digital, and biological capacities, towards the creation of value, in what is called Industry 4.0 or Fourth Industrial Revolution (4IR) [1]. Several studies have shown that Lean Management and IT can match and complement each other [2], 4IR is currently presented as a process of transformation and productive evolution. The globalized behavior of markets, new technological trends and the rise of innovative methodologies have transformed the industry into an interdisciplinary interaction space for organizational decision making [3].

In fact, 4IR is based on new technologies use, production processes and work processes of organizations that have led new businesses and new jobs emergence. This in response to companies' digital transformation that include automation, autonomous and automatic learning, artificial intelligence, internet of things, data analytics, cybersecurity, augmented reality, robotics, nanotechnology, among many other technologies. The rapid growth of 4IR has prompted Education Institutions around the world, to transform current education system into the future education system 4.0 [4]. For this reason, Mexico City Instituto Politecnico Nacional (IPN), as technological education governing institution in this country, had to adapt its training schemes to our society current situation, introducing in its academic programs, cutting-edge topics that provide solutions to current and future problems and that allow the students to be professionally trained. Technology has propelled human civilization further over the past two decades than in the last two hundred years [5]. Education has never been an alien to the great transformations of humanity and even less when they originate in Science and Technology. An example of this are Massive Online Open Courses (MOOCs) and Blended Learning (BL), that are amongst newer technologically enhanced educational approaches that have been introduced. MOOCs and BL are now being implemented to enhance the educational processes within university classrooms worldwide [6], even the English publication entitled the economist carried out an analysis of the current situation in the world and that was strengthened after the pandemic that was experienced in recent years by COVID-19 pandemic [7]. For this reason, it is necessary to make a current reflection

on the academic context and modernize different universities of the world plans and study programs.

1.1 *Instituto Politecnico Nacional*

Mexico City-IPN is the state educational institution created to consolidate, through education, economic, scientific, technological, cultural, and political independence to achieve nation social progress, in accordance with Mexican Revolution historical objectives, which are contained in Mexican United States Political Constitution [8]. Likewise, it is considered the most important technological education institution in Mexico due to the quality of its graduates, their contributions to country development and their social commitment. Mexico City-IPN serves three educational levels such as medium high school level with almost 68,000 students, undergraduate level with almost 135,000 students and postgraduate with almost 4500 students through its different academic units and that serves three branches of knowledge, such as physics and mathematical sciences, administrative and social sciences, and biological and medical sciences. The predominant being is mathematical and physics sciences, and its strength is based in teaching engineering [9]. Its presence in national life, as well as the contributions made for benefit Mexican society, have been possible because Mexico City-IPN has been very clear about the mission it must fulfill and the strategies to face the challenges and opportunities of the national environment. and international where the only constant is change [10] (Table 1).

Since its origins, Mexico City-IPN has been characterized by supporting and promoting Mexican State policies development, related to the industrialization of the country and with the permanent improvement of public, private and social companies. Through preparation among of engineers, masters, and doctors among other professions; executing scientific and technological research projects, with the necessary link to maintain its institutional actions relevance, and with knowledge dissemination that, in its classrooms or with its information technologies, is applied and innovated through educational process. Mexico City-IPN must align and adapt to the national and international context, to become one of the advanced and strategic pillars of the growth and development model with which Mexico, within 4IR framework, will

Table 1 Academic programs in operation and students enrolled 2019–2020 [10]

			Total students		
Description	Men SM	Women SM	Men NSM	Women NSM	Total
Medium high school level	41,619	27,479	754	762	70,614
Undergraduate level	66,670	43,200	1,886	2,319	114,075
Postgraduate level	3,759	2,631	88	86	6,564
Total					191,253

*SM—Scholar Modality, NSM—Non-Scholar Modality

face with justice, sovereignty, independence, nationalism and cosmopolitanism, the challenges and opportunities that arise, for a successful and proactive insertion of the world governance and governability avant-garde modalities [11]. As the governing institution of public technological education in Mexico, is obliged to respond to world needs that require new scientific, technological and innovation capacities. In addition, it is committed to preparing trained people for the social environment and the labor market needs; since it is highly recognized by Mexican society for the role it has played in national development since its inception, and it should preserve it. This social recognition is palpable by the great demand it has from young people to study at our institution. That is why must have a vision of the future, to prepare the next

graduates, since its students will be incorporated into the world of work in 4, 5 or 8 more years, therefore, Mexico City-IPN must strengthen the pertinent aspects that enable student's preparation with these skills, attitudes, knowledge and values, demanded by the new systems of production of goods and services, empowering the student with personalized learning, as well as permanent teacher training [11].

1.2 Education 4.0

The connectivity revolution has brought changes in educational models and teaching paradigms, as the introduction of the internet in everyday life has an impact on the ways of establishing socio-communicative and educational relationships in society. Through the changes that occur thanks to technology, different models of education are established that facilitate the way of learning and understanding Didactics and Pedagogy. Education progressed from 1.0 to 3.0 based on the learning needs of the students: from maintaining a single direction to being bidirectional, having autonomy over content control. For its part, Education 4.0 is based on the main trends of innovation and change. 4IR learnings are focused on XXI century competencies, such as self-direction, self-assessment, and teamwork. Education 4.0 is not an educational model; it is the application of the already existing information technology and communication tools and new technologies generations to prepare people who adapt easily to the changes presented by 4IR. Its main objective is to train future graduates, future professionals, and future technicians to join this new reality of the service industry, where digital technologies play a very important role [11] (Fig. 1).

Some features are:

- Flexible learning and closely linked to productive sector.
- Adaptive learning to each student profile with help of digital media (customization).
- Self-managed learning.
- Learning with constant feedback from student's own progress data analysis (Learning Analytics).
- Cooperation between student and teacher as teaching basis.

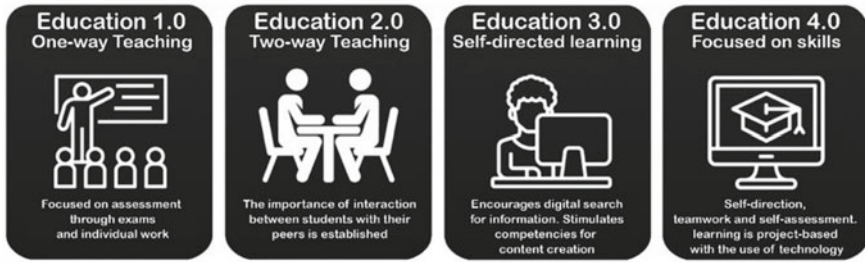


Fig. 1 Education 4.0 evolution [11]

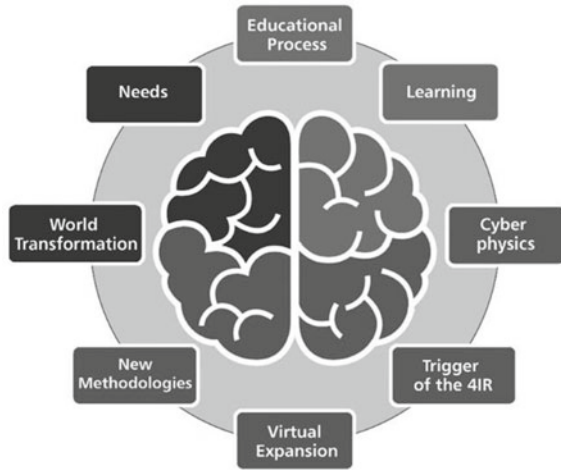
- Communication as the main vehicle for learning.
- Real problem solving is encouraged.
- Incorporates play and real environments creation as learning main objectives.
- Evaluation is a constant process to improve and progress.
- IT use for access, organization, creation, and dissemination of content tool.

In the current digital age, even the designing tasks have been altered and modified to cope with and benefit from the potentials of the new designing methods appeared and made a way of themselves into the conventional design process [12]. In other fields, digital media use has changed the perspective of Industry, Augmented Reality (AR) currently plays an important role to undertake the challenges in integrating technologies to expedite the march towards Industrial Revolution, therefore, the machining industry relies on new design concepts and methods of an innovative human-machine interaction application that overlays virtual components on a real-world environment [13]. It is due to these tools use those new techniques and methodologies based on IT have achieved an evolution of teaching and learning experience in Upper Middle Higher Education and Higher Education, that witnesses many techniques being introduced towards student-centered learning. One of the most popular techniques is game-based learning or known as gamification to enrich students experience in classroom [14]. But to focus on these techniques, we must first define a general strategic transformation.

2 Mexico City-IPN General Strategic Transformation

Instituto Politecnico Nacional as an institution of the Mexican State, has responded to nation needs throughout 83 years, always maintaining its commitment to the principles that gave rise to it. 4IR, characterized by information technologies, communication between machines, robotics, artificial intelligence, internet of things, Big Data, has an impact on practically all disciplines and professions, productive branches, life of societies and governments, Mexico has also undertaken a very profound change aimed at the Fourth Transformation of the nation. In this double environment Mexico

Fig. 2 Strategic transformation agenda [15]



City-IPN must be a relevant part of the change, that is why a transformation of the institution was proposed to ensure its relevance and enhance its capacities to achieve higher levels of quality. The foundations of this transformation have been built with community's participation, in order to make this transformation viable and considering the work carried out, it was necessary to build a Strategic Transformation Agenda that indicates the direction and institutional priorities in the short term [15].

The agenda brings together the results of the work carried out by the community. The strategies, priorities and actions that comprise it have been the product of this collaborative reflection, in addition to being a map that will guide our efforts, the agenda defines growth opportunities through important projects that require intelligence, creativity, and a desire for change. Their results will reinvigorate the identity and pride and have a greater impact on national development (Fig. 2).

The strategic transformation Agenda allows to carry out a transformation in teaching systems for societies benefit, allowing to improve their life quality, so in order to carry out this project it is necessary to get into new technologies use and self-preparation, physical-organizational infrastructure, and financial resources.

2.1 Strategic Actions

Research and Innovation with social impact is about strengthening research, technological development and innovation through the active participation of the community in projects with national impact that affect country's transformation, the way to achieve it, is through institutional policies that encourage, recognize and encourage support staff and students to participate in group research and impact innovation, with their help, it is intended to develop and implement new students profiles, academic staff and support staff will strength talent 4.0 in a new homeland context,

the way in which this is going to be carried out is through an education 4.0 and industry 4.0 specialist team integration with technical, humanistic, socio-cognitive and digital capacities that support teacher's academies in profiles definition and of study programs redesign. The aim is to train Mexico City-IPN student and teachers through education

4.0 with pertinent programs and study plans linked to various sectors with ethical values, pertinent regulations, and the appropriate resources and infrastructure to attend qualified human needs for a more just, inclusive and democratic Mexico according to the 4IR. This will be done through an awareness strategy, information and training for support staff and redesign of study programs. It is necessary to get a needs diagnosis for Infrastructure and equipment 4.0 that supports Mexico City-IPN students training. It will be achieved through a classroom, workshop and laboratory needs survey in every school. Mexico City-IPN governance aims to incorporate more people into academic training offer at its different levels and modalities with inclusion and equity criteria besides excellence (Fig. 3).

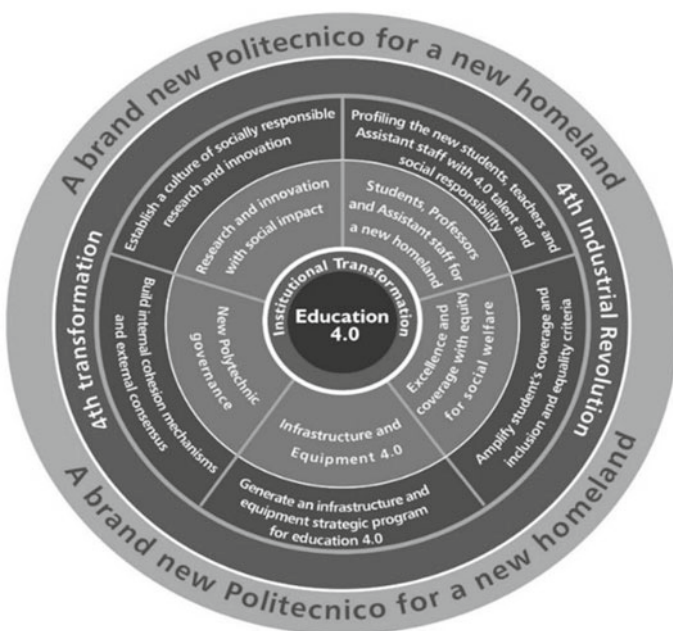


Fig. 3 Transformation General Strategy [15]



Fig. 4 General scheme for curriculum design at Mexico City-IPN [16]

2.2 Mexico City-IPN Methodology for Plans and Programs Redesign Within New Educational and Academic Model Framework

All these new designs followed the methodology proposed in Manual for the plans and programs redesign within new Educational and Academic Model Framework, volume 12 of 2003 Materials for Reform series. In order to design a curriculum, it is necessary to look for answers to several questions: (1) what, when and how to learn and teach, and (2) what, how and when evaluate what was learned and taught. Hence one of the first steps are to define what students are expected to learn and what is going to be taught, what will be the curricular contents in terms of a: necessary knowledge, skills, attitudes, and values to train a professional at Mexico City-IPN. A strategy of successive approximations is proposed to solve the different stages and concentrate the information [16]. This first stage of curriculum design has as purpose to identify essential aspects of the educational project, which it is a part, as well as social demands to which the institution must respond, both institutional as external to the institution, which should be considered to develop the curriculum. In other words, it deals with the rationale and guidelines of curriculum proposal [17] (Fig. 4).

Every stage has important actions to do after identifying educational project, Curriculum referents like Mission, Vision, Educational Model, External references, public policies and environmental diagnosis, then Curricular objectives and profiles as the General objective of the program and professional field, entry profile, graduation profile. academic model and curricular objectives. Study Plan Design like Content selection and content research, Course design, Subjects design and pedagogical and didactic guidelines and finally Implementation of the study plan, Teacher's profile and academic person training, operation conditions development. The proposed methodology follows the canons of curriculum design, but with an adaptation based on strategic planning and participatory that has been used in Academic Reform process. In order to develop each stage mentioned above, it will be essential to use the SISPLAN software that was developed by National Computing Center for CourseWorkshop "Strategic planning of structural change and curriculum in Mexico CityIPN. This software is a tool that allows to identify ideas and opinions around topics defined, its use facilitates agile and harmonious participation of a working group, this is achieved through the exposition of approaches, the consideration of their input and a review assessment of said contributions [16]. The proposal seeks to provide teachers, who are the builders of curricular designs, with the information and guidance necessary for the development of flexible programs focused on

learning in accordance with the Institute's educational proposal. For this reason, the design by competencies is incorporated, that is, the knowledge, skills, attitudes that are put into play in an integrated manner for performance in a field of action [18], also, it is based on a constructivist perspective of learning that considers that knowledge is built from the re-elaboration of the previous knowledge of the learner, from the internal reorganization of schemes and from the conflict between what is known and what should be known [19]. The results of the analysis of the internal referents and external sources provide essential information to delimit the profile of the graduate, the curricular objectives and some of the criteria necessary to determine the content curriculum (knowledge, skills, attitudes, and values). When it comes to designing a new program educational, this part of the curricular design has the purpose, in addition to the above, to assess your viability.

3 Discussion and Results

The first strategic action to achieve Education 4.0 guiding priority is based on new talent development through education 4.0 at different educational levels offered at Mexico City-IPN (medium high, undergraduate and postgraduate level), it's current situation does not have the programmatic or structural normative elements to effectively meet 4IR environment needs, so it is necessary to carry out a strategy based on a road map for education 4.0 implementation that has a proposal for regulatory requirements, an awareness program, a preliminary draft of financial self-determination, and a bonding strategy to present updated study plans and programs designed with 4.0 elements. It is necessary to create an entrepreneurial and research culture in the community to innovate and influence the transformation of the country, but the support and efforts to implement relevant programs to promote and create an entrepreneurial culture are unlimited and this slows down the momentum of the innovation, the development of new products, services and the training of highly competitive professionals, which is why a strategy is necessary that articulates the potential and institutional strengths to generate an entrepreneurial culture with the use of technology and ingenuity to serve society.

Another necessary action is to develop new access, education, and training programs talent 4.0, but many people in Mexico CityIPN are unaware of education 4.0 context and their graduates lack of training in industry 4.0 skills required and do not have teaching and technological skills focused on education 4.0. To achieve this, it is necessary to integrate a group of specialists in education 4.0 and in industry 4.0 who will carry out an analysis of industry 4.0 in Mexico development to define the required professional profiles and the definition of the teachers who will train the 4.0 talent, to increase access to education with equity and inclusion principles is necessary to increase physical and academic infrastructure for new social demands. It is important to mention that there is an absence of an inclusive culture and specialized human resources, especially one that is aimed at admission processes because do not have a comprehensive care program for people from highly vulnerable sectors, and

there are insufficient learning unit's different modalities offered without mentioning that there is no academic flexibility. Therefore, it is necessary to create a strategic plan for infrastructure, equipment and institutional services by dependency. Unfortunately, there is still no precise information on the needs to attend to these academic programs in accordance with education 4.0.

4 Conclusions

There is no clarity on the concept of Education 4.0, Industry 4.0 and Economy 4.0 by students and teachers, which builds the way in which they perceive the new designed plans and programs, due to change resistance.

Education 4.0 also known as smart education, is a transformation of education to solve new human needs based on 4IR, this allows CIT's technologies incorporation, as well as artificial intelligence, data analytics, education gamification portability among other techniques and apply them to all educational models through flexible, adaptive, self-managed, and feedback learning.

All of this based on cyber-physical systems such as, IoT to project technologies such as, robotics, mechatronics, Big Data, nanotechnology, among others. To achieve this virtual expansion, it is necessary to carry out education in virtual spaces for academic and work purposes according to community needs using new methodologies such as e-learning, m-learning, b-learning, among others.

Education 4.0 is not an educational model, it is an adaptive teaching and learning process, therefore both students and teachers must be trained in the use of technological tools to implement them in classroom.

The current organizational structure by careers must tend towards inter-disciplinarity like the new ones to be part of an education 4.0 approach, and the contribution of the community to achieve transformation requires the collaborative work of teachers and students, through monitoring and compliance with the general transformation strategy.

There is Complexity to acquire adequate infrastructure for Education 4.0 careers laboratories due acquisition policies, but it is necessary great investment in Connectivity and infrastructure in every Mexico City-IPN school, besides government and internal regulations need to be modified in order to accomplish new strategic agenda objectives.

It is necessary to strengthen the link with productive sectors of goods and services to know their trends and Talent 4.0 requirements, as well as know the infrastructure used in their processes because it is necessary to establish alliances in research, development, and innovation projects, including technology transfer to improve the infrastructure of Mexico City-IPN.

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